cycle. From this point of view, both the inflationary model and the cyclic model rely on accelerated expansion—the inflationary model near its beginning and the cyclic model at the end of each of its cycles—but only the latter has direct observational support. (Remember, the cyclic approach is designed so that we are just entering the trillion-year phase of accelerated expansion, and such expansion has been recently observed.) That's a tick in the cyclic model's column, but it also means that should accelerated expansion fail to be confirmed by future observations, the inflationary model could survive (although the puzzle of the missing 70 percent of the universe's energy budget would emerge anew) but the cyclic model could not.

New Visions of Spacetime

The braneworld scenario and the cyclic cosmological model it spawned are both highly speculative. I have discussed them here not so much because I feel certain that they are correct, as because I want to illustrate the striking new ways of thinking about the space we inhabit and the evolution it has experienced that have been inspired by string/M-theory. If we are living within a three-brane, the centuries-old question regarding the corporeality of three-dimensional space would have its most definite answer: space would be a brane, and hence would most definitely be a something. It might also not be anything particularly special as there could be many other branes, of various dimensions, floating within string/M-theory's higher dimensional expanse. And if cosmological evolution on our three-brane is driven by repeated collisions with a nearby brane, time as we know it would span only one of the universe's many cycles, with one big bang followed by another, and then another.

To me, it's a vision that's both exciting and humbling. There may be much more to space and time than we anticipated; if there is, what we consider to be "everything" may be but a small constituent of a far richer reality.